Multilevel Heterogeneous Domain Adaptation Method for Remote Sensing Image Segmentation

Chenbin Liang, Bo Cheng, Baihua Xiao, Yunyun Dong, and Jinfen Chen

***Abstract*— Due to more abundant data sources, more various objects of interest, and more time-consuming annotations, there is a large amount of out-of-distribution (OOD) data in the remote sensing field, on which the performance of high-accuracy image segmentation models trained under ideal experimental conditions generally degrades dramatically. Domain adaptation (DA) consequently comes into being, which aims to learn the predictor for the label-scarce target domain of interest with the help of the label-sufficient source domain in the presence of the distribution difference, namely, domain shift, between the two domains. However, the off-the-shelf DA methods for image segmentation not only struggle to cope with the more complex domain shift problems in remote sensing imagery but also almost cannot process heterogeneous data directly without information loss. While the current heterogeneous DA methods mostly still rely on some supervision information from the target domain, which is typically inaccessible in the real world. To overcome these drawbacks, we propose the multilevel heterogeneous unsupervised DA (UDA) method, termed MHDA, which unifies the instance-level DA based on cycle consistency, the feature-level DA based on contrastive learning, and the decision-level DA based on task consistency into a framework to more effectively handle the complex domain shift and heterogeneous data. After that, extensive DA experiments are conducted on the International Society for Photogrammetry and Remote Sensing (ISPRS) dataset, the BigCity dataset constructed by ourselves, and the Wuhan University (WHU) dataset, to explore the effect of each module in MHDA, the necessity of heterogeneous DA, and the effectiveness of multilevel DA. And the results demonstrate that MHDA can achieve superior performance on the remote sensing image segmentation task, compared with several state-of-the-art DA methods.**

***Index Terms*— Contrastive learning, cycle consistency, domain adaptation (DA), semantic segmentation, task consistency.**

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I. Introduction

FROM space-based satellite to air-based unmanned aerial vehicle (UVA) to ground-based radar, the space–air–ground integrated network for remote sensing data acquisition has been established, which promotes the explosive growth of its data volume, significantly expands observation range, and markedly enriches observable objects. Various disciplines attempt to use remote sensing data to monitor their objects of interest in real-time, and a variety of sensors have emerged one after another for different applications, such as the sensor with the low spatial resolution and 36 spectral bands (0.4–14 μm) in the MODIS satellite for meteorological observation, the sensor with the medium spatial resolution and 11 spectral bands (0.4–12.5 μm) in the Landsat 8 satellite for terrestrial ecosystem monitoring, and the sensor with the high spatial resolution and four spectral bands (0.4–0.9 μm) in the GaoFen-2 satellite for urban objects monitoring. Multiapproach data acquisition and a wealth of sensors not only facilitate its wide applications but also determine its multisource heterogeneous properties. Therefore, the professional and fine interpretation of multisource heterogeneous remote sensing imagery plays an important role in fully using data and gaining insight into objects of interest in each professional field. And the rapid development of pixel-level image segmentation technologies in the computer vision (CV) field provides the possibility for its implementation.

With the successful advances of deep learning, convolutional neural network (CNN)-based image segmentation models have become more and more prominent in many challenges and public datasets. However, when launching a high-accuracy segmentation model into practical scenarios, the performance is generally unstable and unreliable. Zhang et al. [1] summarized that the three basic and implicit assumptions followed by these deep learning models during training restrain their performance in the real world.

1) **Independent and identically distributed (i.i.d.) assumption.** It assumes that the samples are independent of each other, and the training and test data are identically distributed.
2) **Closed-world assumption.** It assumes that the number of categories in the output space and the type of patterns in each category are both fixed, all the samples come from these fixed categories, and each category is also derived from these fixed patterns.
3) **Clean and big data assumption.** The training data are assumed to be precisely annotated, and the data

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volume is assumed to be large enough to cover different variations. And obviously, the other two assumptions are sufficient conditions for the i.i.d. assumption, given that when the real tasks violate either of the other two assumptions, there will certainly be the distribution difference between the training and test data.

Whereas in the remote sensing image segmentation task, it will face more challenges if the model trained under ideal experimental conditions is to be extended to the real world for the following reasons.

1) **More complex sensors.** Various sensors receive data with diverse spatial and spectral resolutions. If the training and test data come from different sensors, it will inevitably violate the i.i.d. assumption.
2) **More variable target objects.** Numerous considerations, such as the vast monitoring range, diverse geographical environment, changeable climatic conditions, and different phenological periods, make each category in remote sensing imagery generally have more varied patterns, which makes it more difficult to satisfy the closed-world assumption.
3) **More complicated annotation work.** In view of the particularity of the objects in remote sensing, their pixel-level interpretation generally requires more expert knowledge and more expensive time costs, which makes it more difficult to follow clean and big data assumption.

The complexity of the data itself, the variable objects of interest, and the time-consuming annotation work all put forward strong requirements for breaking the three basic and implicit assumptions in the remote sensing image segmentation task.

Recently, to smash the shackles brought by these assumptions, lots of related studies have been launched. As one of the successful attempts to break the core i.i.d. assumption, domain adaptation (DA) [2] has been widely concerned and aims to execute the target domain task with the label-scarce target domain and the label-sufficient source domain in the presence of domain shift between the two domains. The current DA methods can be categorized into supervised DA, semi-supervised DA, and unsupervised DA (UDA) based on the presence or absence of the labeled target domain data, into homogeneous DA and heterogeneous DA according to whether the two domains have the same dimension space, and into instance-level DA, feature-level DA and decision-level DA based on the objects used for domain alignment. At present, the related literature in the CV field mostly focuses on homogeneous unsupervised DA, and the level of aligned objects in the specific task is generally subject to its own advantages and disadvantages. For example, feature-level DA has shown remarkable performance on the coarse-grained CV tasks [3], [4], [5], such as image classification and object detection, while in view of the demand for fine-grained domain alignment, instance-level DA and decision-level DA are more favored by image segmentation [6], [7], [8]. However, unlike the natural images in the CV field, remote sensing imagery is confronted with the more complex domain shift problems brought by wider observation range and more variable observable objects. Therefore, it is inappropriate to directly apply the off-the-shelf DA methods for image segmentation to the remote sensing field, and the more effective DA method for the remote sensing image segmentation task needs to be further investigated.

Nevertheless, with the exception of more complex domain shift problems, there is also a large number of heterogeneous data caused by various spectral resolutions in remote sensing. Unfortunately, homogeneous unsupervised DA methods are nearly incapable of directly processing heterogeneous data. Although some research [9], [10], [11] involves heterogeneous remote sensing images, they have to artificially remove some bands to make the two domain data homogeneous in DA experiments, which unavoidably results in some information loss and may even be detrimental to some category extraction. However, in contrast to the successful advances of homogeneous unsupervised DA, where the supervision information is only from the source domain, the current heterogeneous DA methods generally require paired samples [12], [13], [14] in the two domains or a certain amount of labeled data [15], [16], [17], [18] in the target domain to bridge the heterogeneous domains, and the transformation functions used to eliminate heterogeneity typically adopt shallow learning models with limited ability to handle complex scenarios. Besides, although several literature has pursued research [19], [20], [21] on heterogeneous unsupervised DA, they basically only use feature-level constraints that are inadequate for fine-grained domain alignment. Therefore, there is also an urgent requirement for the DA method that can completely exploit heterogeneous data information in the remote sensing image segmentation task.

To address the above issues, we propose a multilevel heterogeneous DA method, termed MHDA, for the remote sensing image segmentation task. On one hand, MHDA unifies the three-level DA into a framework for collaborative training to achieve their complementary advantages, which not only satisfies the requirement for fine-grained domain alignment in the image segmentation task but also obtains the domain-invariant features with better generalization ability to cope with the more complex domain shift problems in remote sensing imagery. On the other hand, MHDA uses a deep learning model for unsupervised unpaired heterogeneous data transformation and simultaneously imposes feature-level and decision-level constraints, which contributes to effective heterogeneous unsupervised DA. Fig. 1 sketches a visualization of the flowchart of MHDA and briefly clarifies the main technologies and functions of each-level DA.

In the following, we first survey the advantages and disadvantages of each-level DA and review the current heterogeneous DA methods in Section II. And then, in Section III, we describe each module in MHDA and objective functions in detail. After that, in Section IV, we choose the International Society for Photogrammetry and Remote Sensing (ISPRS), BigCity, and Wuhan University (WHU) datasets to conduct extensive DA experiments and analyze their main domain shift problems, then investigate the effect of each module in MHDA through ablation studies, and then make comparative experiments to explore the necessity of heterogeneous DA and the effectiveness of multilevel DA. Overall, the main contributions can be summarized as follows.

1) Realize an instance-level domain alignment module with cycle consistency (IDAC). It directly performs the unsupervised unpaired image translation between

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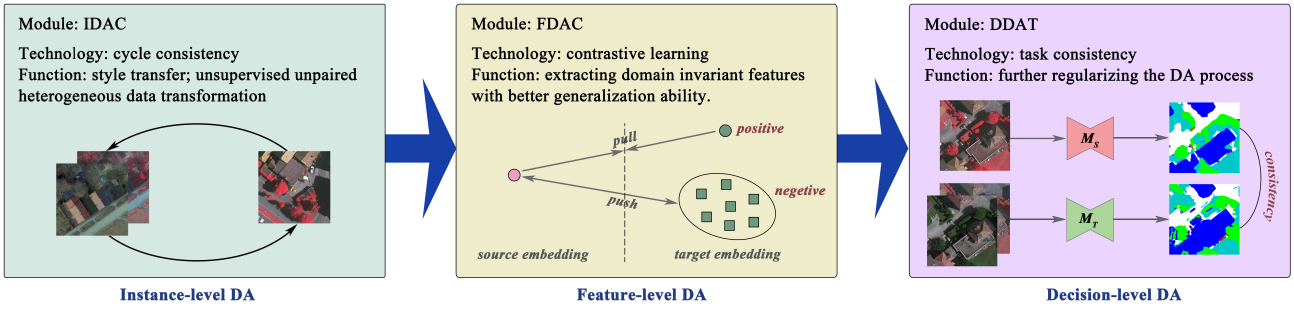


Fig. 1. Flowchart of MHDA. After implementing instance-level DA, feature-level DA, and decision-level DA in turn, MHDA can effectively handle the complex and fine-grained domain shift problems and fully use heterogeneous data information.

heterogeneous remote sensing images with different spectral resolutions by the use of the deep learning model, CycleGAN [22] and lays a foundation for further mining the additional band information in the follow-up.

2) Design a feature-level domain alignment module with contrastive learning (FDAC). Given that contrastive learning can allow the model to learn generic features with better generalization ability, it is introduced to extract more effective domain-invariant features for feature-level domain alignment.
3) Propose a decision-level domain alignment module with task consistency (DDAT). It is used to further regularize the DA process and argues that the prediction results of the data in one domain should be consistent with those of its corresponding translated data in the other domain.
4) Unify the three-level DA into a framework for their complementary advantages, which not only enables it to directly handle heterogeneous data but also allows it to tackle the more complex and fine-grained domain shift problems in the remote sensing image segmentation task. And it outperforms other state-of-the-art DA methods in each DA experiment.

II. Related Work

A. Notations

A domain $\mathcal{D}$ consists of the data X and its distribution $P(x)$, i.e., $\mathcal{D} = \{X, P(x)\}$, $x \in X$. A task $\mathcal{T}$ is composed of the label space Y and prediction function $f(x)$, where $f(x)$ can be regarded as the posterior probability $p(y|x)$, i.e., $\mathcal{T} = \{Y, P(y|x)\}$, $y \in Y$. The domain with sufficient labeled data is the source domain $\mathcal{D}_S$ and its task is $\mathcal{T}_S$. The domain of interest is the target domain $\mathcal{D}_T$ and its task is $\mathcal{T}_T$.

In the context of DA, it assumes that the source and target domains have the same task (i.e., $\mathcal{T}_S = \mathcal{T}_T$) but distinct distributions (i.e., $P(x_s) \neq P(x_t)$, where $x_s \in X_S$ and $x_t \in X_T$. Especially, in homogeneous DA, $X_S = X_T$, and in heterogeneous DA, $X_S \neq X_T$), and there are sufficient labeled samples in the source domain, while the label space Y_T in the target domain is scarce. Concretely, both the supervised DA and semi-supervised DA assume that there is a certain amount of labeled data in the target domain, and the difference is that the former only uses the labeled samples in the two domains, while the latter further uses unlabeled data in the target domain as well. While unsupervised DA assumes that the label space of the target domain Y_T is entirely unseen.

B. Definition

Given the labeled source domain $\mathcal{D}_S$ and the unlabeled target domain $\mathcal{D}_T$, where $X_S \neq X_T$ and $P(x_s) \neq P(x_t)$, the goal of heterogeneous unsupervised DA is to develop a model $f : x \to y$ that can perform well on $\mathcal{D}_T$, by the use of known X_S, Y_S, and X_T.

In this article, we aim to propose an effective heterogeneous unsupervised DA method for remote sensing image segmentation, which can fully use the heterogeneous data information and handle the complex and fine-grained domain shift problems. As stated above, DA can be performed by aligning objects at the instance, feature, or decision level, depending on the specific task. While both the current DA methods for image segmentation and heterogeneous DA methods cannot seamlessly extend to our goal. To this end, in this section, we carefully survey and analyze the advantages and disadvantages of each-level DA, as well as review and explore the related work on heterogeneous DA in detail.

C. Instance-Level DA

The basic idea of instance-level DA is to use the real labeled sample (x_s, y_s) in the source domain to generate the pseudolabeled sample $(A(x_s), y_s)$ in the target domain, where $A(\cdot)$ is seen as a way of simulating domain shift. Based on how $A(\cdot)$ is formulated, there are three main methods.

1) **$A(\cdot)$ is one or more traditional data augmentation functions.** This approach has already been shown to be very effective in DA methods [23], [24], [25] for medical image analysis tasks. And it has also been extensively discussed in the CV field. For example, Volpi and Murino [26] formulated a combinatorial optimization strategy of data augmentation, and Shi et al. [27] also synthesized several semantically meaningful variations using traditional augmentation functions.
2) **$A(\cdot)$ is a function for constructing adversarial examples.** Inspired by adversarial attacks, some researchers use the worst case perturbations to simulate the domain shift. They generally synthesize adversarial examples using adversarial gradients from the task classifier [28] or the domain classifier [29] for DA training.
3) **$A(\cdot)$ is a neural network.** Especially with the advances of style transfer technology, many unsupervised unpaired image-to-image translation models, such as CycleGAN [22], DiscoGAN [30], and DualGAN [31], have been successively introduced into the DA field.

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And through these image translation models, some literature [9], [32] attempts to obtain the generated images from the source domain with the same data distribution to the target domain, and vice versa.

In conclusion, instance-level DA uses various $A(\cdot)$ to simulate the domain shift between the two domains at the pixel level, which assists with fine-grained domain alignment, but also has the obvious disadvantage of overreliance on generated data, thus inducing more robustness issues. For instance, in digit recognition, using the horizontal/vertical flipping operation as $A(\cdot)$ will readily cause label shift, while adopting a neural network as $A(\cdot)$ may even generate interference that is too sophisticated to detect. Therefore, although instance-level DA is a popular method in image segmentation due to the pixel-level alignment, if only implementing instance-level DA, it will be easy to have the opposite effect.

D. Feature-Level DA

As the initial attempt, feature-level DA has been broadly and intensively discussed. Some loss functions are used to directly constrain features to be domain-invariant. The earliest work adopts some traditional distance metrics, such as maximum mean discrepancy (MMD) [3], joint MMD (JMMD) [4], Kullback–Leibler (KL) divergence [33], and Wasserstein distance [34]. And with the development of generative adversarial networks (GAN) [35], adversarial learning is also injected into feature-level DA. The basic idea is to implement one or more domain classifiers [5], [36] to distinguish whether the features come from the source domain or the target domain and realize feature-level domain alignment based on the adversarial loss. Actually, Goodfellow et al. [35] also demonstrated that the learning goal of GAN can be equivalent to minimizing the Jensen–Shannon (JS) divergence between the real and generated data distributions, which provides the theoretical support for domain alignment via adversarial learning as well.

Except for direct alignment with cost functions, constructing the domain-invariant information is also an effective way of feature-level DA, mostly inspired by the autoencoder (AE) [37]. As a typical unsupervised learning algorithm, AE, composed of the encoder and decoder, aims to achieve the reconstruction of the input. A lot of literature has brought AE and its variants into the DA field to realize indirect feature-level domain alignment. Some studies [38], [39] design distinct reconstruction tasks to make the coded information domain-invariant. Other literature [40], [41] uses several adaptive weight strategies of the encoder for domain alignment.

Besides, self-supervised learning has also attracted much attention. It allows a model to learn the generic features regardless of specific tasks by the use of free labels generated from data itself. It can be roughly broken down into two classes: the generative and contrastive methods. The generative method trains DNN to execute pretext tasks [42], [43], such as puzzles, image inpainting, and colorization, for generic feature extraction. The contrastive method [44], [45] aims to learn transformation-invariant features, which first generates multiple patterns of the same instance by different data augmentation functions, then maps them as multiple embeddings with DNN, and uses contrastive loss to bring the embeddings from the same instance closer and separate the embeddings from different instances. In recent years, some researchers [46], [47], [48] have successfully extracted the domain-invariant features via self-supervised learning, which either use generative method for indirect alignment or adopt contrastive loss to directly narrow the distribution distance.

In conclusion, feature-level DA maps the data from the two domains into high-dimensional feature spaces for direct or indirect distribution alignment. Although it is not as intuitive and fine as instance-level DA, it can help features with better generalization ability, and hence it is widely used in many coarse-grained CV tasks. But obviously, this coarse-grained domain alignment in high-dimensional spaces is insufficient for the fine-grained segmentation task. Thus, if only feature-level DA is adopted, the DA effect is generally limited.

E. Decision-Level DA

Since $\mathcal{T}_S = \mathcal{T}_T$, decision-level DA thinks Y_S and Y_T with the same distribution and realizes DA through aligning the output spaces. Actually, the constraints provided by decision-level DA can also be used to potentially promote instance-level or feature-level domain alignment.

In supervised/semi-supervised DA, decision-level DA is the most commonly used strategy [49], [50], [51], which typically implements a domain-shared classifier with label information from the two domains. In unsupervised DA, researchers generally achieve pixel-level or class-aware alignment in the output spaces through some loss functions, such as adversarial loss [6] and mean squared error [52]. Besides, inspired by self-training [53], which is a typical semi-supervised learning method, other studies [54], [55], [56] attempt to generate pseudolabels of the unlabeled target domain data for classifier adaptation to align the two domains at the decision level.

For the image segmentation task, the advantage of decision-level DA is that it can realize fine-grained domain alignment and implicitly promote feature-level or instance-level DA. The disadvantage is that when the real label space of the target domain (Y_T) is unseen, there will inevitably be some noise interference that has a negative impact on the DA effect.

F. Heterogeneous DA

At present, most related studies focus on heterogeneous supervised/semi-supervised DA, which can be divided into two types according to the way of eliminating heterogeneity.

1) Projecting Data From One Domain to the Other: This method generally uses shallow learning models as the transformation function, which cannot tackle complex scenarios. Therefore, these researchers [16], [17], [18], in addition to the instance-level marginal distribution alignment, generally have to align the conditional distribution at the instance level based on the label information from the two domains. Besides, some literature [17], [18] further performs classifier adaptation for improvement at the decision level as well.

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2) Finding a Domain-Invariant Subspace: This method uses domain-specific transformation functions to project the two domains into a domain-invariant common subspace. The related literature [57], [58], [59] generally learns these transformation functions by the domain-shared classifier with reliance on the labeled samples from the two domains. And for further improvement, some studies not only implement classifier adaptation but also achieve feature-level constraints for this subspace using paired samples [12], [14] or some loss functions [15], [60], such as MMD and adversarial loss.

In comparison, heterogeneous unsupervised DA methods are rarely studied and typically solely adopt feature-level DA. Li et al. [19] used MMD and local consistency evaluated by cosine similarity as feature-level constraints to obtain the domain-invariant space. Shared fuzzy equivalence relation (SFER) [20] projects the two domain data into an n-dimensional fuzzy space, measures the similarity of two fuzzy vectors, and builds the fuzzy equivalence relations matrix of each domain for feature-level domain alignment. Feng et al. [21] constructs homogeneous representations by minimizing the distance metric based on constraints associated with linear monotonic maps (LMMs). Besides, although there are several studies [13], [61] not requiring the labeled samples from the target domain, they still rely on other additional information, such as paired samples and common channels, to bridge the source and target domains.

In conclusion, the most discussed heterogeneous supervised/semi-supervised DA methods require label information from the target domain for classifier adaptation or conditional distribution alignment and generally use shallow learning models as transformation functions. Although there are several studies on heterogeneous unsupervised DA, they generally only use feature-level DA, which cannot effectively achieve fine-grained domain alignment. Due to these drawbacks, the above methods cannot directly serve as the heterogeneous unsupervised DA method for the remote sensing image segmentation task.

G. Motivation

At present, there has been some effective research on homogeneous unsupervised DA for the image segmentation task. These methods mostly implemented instance-level DA and decision-level DA [6], [7], [8], and some researchers also used feature-level DA as an auxiliary item for further improvement [62]. Several typical methods are as follows.

1) UDA [9]: Proposed in 2019 and only using instance-level DA, it fine-tunes the image segmentation network pretrained by the labeled source domain data with the pseudolabeled samples in the target domain, which are synthesized by translating the labeled samples in the source domain into the target domain through CycleGAN.

2) AdaptSegNet [6]: Proposed in 2018 and only adopting decision-level DA, it implemented domain classifiers to determine whether the prediction results came from the source domain or the target domain at pixel level and realized the output spaces alignment through the adversarial loss.

3) CyCADA [62]: Proposed in 2018 and using the combination of instance-level DA and feature-level DA, it uses CycleGAN for instance-level domain alignment, implements a domain classifier to determine the features whether came from the source domain or the target domain, and simultaneously maintains the cycle consistency of features.

Although these homogeneous unsupervised DA methods have made some achievements, they cannot process heterogeneous data directly without information loss, and the domain shift problems in remote sensing imagery are more complex as well. In this article, we first implement a deep learning model, CycleGAN, at the instance level for unsupervised unpaired heterogeneous data transformation. However, it is far from enough to handle the more complex domain shift, given that instance-level DA is vulnerable to the noise interference in generated data. Therefore, we also use the contrastive loss and task consistency for decision-level and feature-level domain alignment, respectively, and further unify the three-level DA into a framework for their complementary advantages. Then, the proposed method MHDA can realize effective heterogeneous unsupervised DA for the remote sensing image segmentation task.

III. Methodology

Fig. 2 illustrates the framework of the proposed multilevel heterogeneous DA method, termed MHDA. The segmentation model in the source domain M_S is composed of the encoder F_S and the decoder U_S, i.e., $M_S = U_S \circ F_S$, and similarly there is $M_T = U_T \circ F_T$ in the target domain. $G_{S\to T}$ is the generator for translating the images in the source domain into the target domain, and $G_{T\to S}$ is the reverse generator. DR_S and DR_T are the feature dimension reduction modules for the source and target domains respectively. Altogether, there are three-level DA modules in MHDA, which are incorporated into a framework for collaborative training.

A. Instance-Level Domain Alignment Module With Cycle Consistency

Uses CycleGAN as the unsupervised unpaired heterogeneous data transformation function to achieve instance-level domain alignment with cycle consistency. To be concrete, given a set of inputs (x_s, x_t), IDAC can generate different domain patterns of the same instance, where the source and target domain patterns of x_s are itself and $x_{s\to t} = G_{S\to T}(x_s)$, respectively, and accordingly, those of x_t are $x_{t\to s} = G_{T\to S}(x_t)$ and itself, respectively.

B. Feature-Level Domain Alignment Module With Contrastive Learning

Uses the contrastive loss for direct feature-level domain alignment. Different domain embeddings of the same instance can be generated by feeding the different domain patterns of the same instance obtained in IDAC into the domain-specific encoders (F_S and F_T). The contrastive loss is used

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to bring the source domain embedding of one instance closer to its corresponding target domain embedding, and farther away from the target domain embeddings of other instances, which enables the domain-invariant features extracted by the encoders to address the more complex domain shift problems in remote sensing imagery.

C. Decision-Level Domain Alignment Module With Task Consistency

Realizes fine-grained decision-level domain alignment with task consistency. Given the assumption of $\mathcal{T}_S = \mathcal{T}_T$, the prediction results of different domain patterns of the same instance should be in line with each other. Therefore, we propose the task consistent loss that can further constrain the overall framework at the decision level.

D. Parameter Update and Objective Function

Different modules in MHDA adopt different parameter update methods. First, F_T (except for the first convolutional layer) and DR_T are frozen, and other modules are updated by back-propagation with $\mathcal{L}$, as follows:

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}} + \lambda_{\text{att}}\mathcal{L}_{\text{AttenNCE}} + \lambda_{\text{rec}}^{\text{seg}}\mathcal{L}_{\text{rec}}^{\text{seg}} \tag{1}$$

where λ_{cyc}, λ_{att}, and $\lambda_{\text{rec}}^{\text{seg}}$ are hyperparameters; $\mathcal{L}_{\text{seg}}$ is the domain-specific semantic segmentation loss; $\mathcal{L}_{\text{cyc}}$ is the overall object function of CycleGAN for the instance-level cross domain constraint; $\mathcal{L}_{\text{AttenNCE}}$ is the contrastive loss for the feature-level cross domain constraint, which further encourages the encoders to extract the domain-invariant features with better generalization ability; $\mathcal{L}_{\text{rec}}^{\text{seg}}$ is the task consistent loss for the fine-grained cross domain constraint at the decision level.

After that, denoting the parameters of F_T (except for the first convolution layer) and DR_T as θ_T, and those of F_S (except for the first convolution layer) and DR_S as θ_S, we momentum update θ_T as follows:

$$\theta_T \leftarrow \lambda_\theta \theta_T + (1 - \lambda_\theta)\theta_S \tag{2}$$

where $\lambda_\theta \in [0, 1)$ is the momentum coefficient.

E. Domain-Specific Semantic Segmentation Loss $\mathcal{L}_{seg}$

The cross-entropy loss is used as our semantic segmentation loss, where the segmentation loss in the source domain can be expressed as (3). And combining y_s with $x_{s\to t}$ to form the pseudolabeled sample in the target domain, the segmentation loss in the target domain can be formulated as (4)

$$\mathcal{L}_{\text{seg}}(X_S, Y_S; M_S) = -\mathbb{E}_{x_s\sim X_S, y_s\sim Y_S}\sum_{i=1}^{N} y_s^i \log M_S(x_s)^i \tag{3}$$

$$\mathcal{L}_{\text{seg}}(X_S, Y_S; G_{S\to T}, M_T) = -\mathbb{E}_{x_s\sim X_S, y_s\sim Y_S}\sum_{i=1}^{N} y_s^i \log M_T(x_{s\to t})^i \tag{4}$$

where N is the number of categories.

F. Instance-Level Cross Domain Constraint $\mathcal{L}_{cyc}$

CycleGAN, an unsupervised unpaired image translation model, is used for the instance-level domain alignment and heterogeneous data transformation. The overall objective function $\mathcal{L}_{\text{cyc}}$ (5) consists of the domain-specific adversarial loss $\mathcal{L}_{\text{adv}}$ and the reconstruction loss $\mathcal{L}_{\text{rec}}^{\text{img}}$

$$\mathcal{L}_{\text{cyc}} = \mathcal{L}_{\text{adv}} + \lambda_{\text{rec}}^{\text{img}}\mathcal{L}_{\text{rec}}^{\text{img}}. \tag{5}$$

$\mathcal{L}_{\text{adv}}$ is used to align the real and synthetic data distributions. Two discriminators D_S and D_T are deployed to determine whether the data come from the source domain or the target domain. Specifically, D_S addresses to align the distributions of x_s and $x_{t\to s}$, and the adversarial loss in the source domain can be defined as follows:

$$\mathcal{L}_{\text{adv}}(X_S, X_T; G_{T\to S}, D_S) = \mathbb{E}_{x_s\sim X_S}\left[\log D_S(x_s)\right] + \mathbb{E}_{x_t\sim X_T}\left[\log(1 - D_S(x_{t\to s}))\right]. \tag{6}$$

Similarly, there is also the adversarial loss in the target domain $\mathcal{L}_{\text{adv}}(X_T, X_S; G_{S\to T}, D_T)$.

Besides, to maintain the image structure invariance in unsupervised unpaired heterogeneous data transformation, CycleGAN argues that the learned transformation functions should be cycle-consistent: $G_{T\to S}(G_{S\to T}(x_s)) \approx x_s$ and $G_{S\to T}(G_{T\to S}(x_t)) \approx x_t$, and $\mathcal{L}_{\text{rec}}^{\text{img}}$ is used to ensure this cycle consistency, as follows:

$$\begin{aligned}\mathcal{L}_{\text{rec}}^{\text{img}}(X_S, X_T; G_{S\to T}, G_{T\to S}) &= \mathbb{E}_{x_s\sim X_S}[\|G_{T\to S}(G_{S\to T}(x_s)) - x_s\|_1] \\ &\quad + \mathbb{E}_{x_t\sim X_T}[\|G_{S\to T}(G_{T\to S}(x_t)) - x_t\|_1].\end{aligned} \tag{7}$$

G. Feature-Level Cross Domain Constraint $\mathcal{L}_{AttenNCE}$

In this article, we use the improved contrastive loss to extract domain-invariant features. As shown in Fig. 2, we pass different domain patterns of x_s into the domain-specific encoders (i.e., x_s to F_S and $x_{s\to t}$ to F_T) and concatenate high-level and low-level features after dimension reduction with the domain-specific modules, DR_S and DR_T, to obtain the source and target domain embeddings of x_s, namely e_s and $e_{s\to t}$. Consider $e_{s\to t}$ as the positive sample of e_s, and $\{q_1, q_2, \ldots, q_r\}$ as the negative samples. These negative samples are kept in a queue known as the *memory bank*, which is composed of the target domain embeddings of other previously processed instances. And after the current calculation, the top element pops from the front of *memory bank* and $e_{s\to t}$ enters from the rear. With similarity measured by the dot product and NCE losses, the contrastive loss $\mathcal{L}_{\text{InfoNCE}}$ [44] can be calculated as follows:

$$\mathcal{L}_{\text{InfoNCE}} = -\log\left(\frac{\exp(e_s \cdot q_0/\tau)}{\sum_{i=0}^{r}\exp(e_s \cdot q_i/\tau)}\right) \tag{8}$$

where τ is a temperature hyperparameter, and for simplicity, $e_{s\to t}$ is rewritten as q_0.

However, it is worth noting that these negative samples are from different instances, but not necessarily from different semantic categories, which means that they are not the negative samples of e_s in the strict sense. If these freely generated

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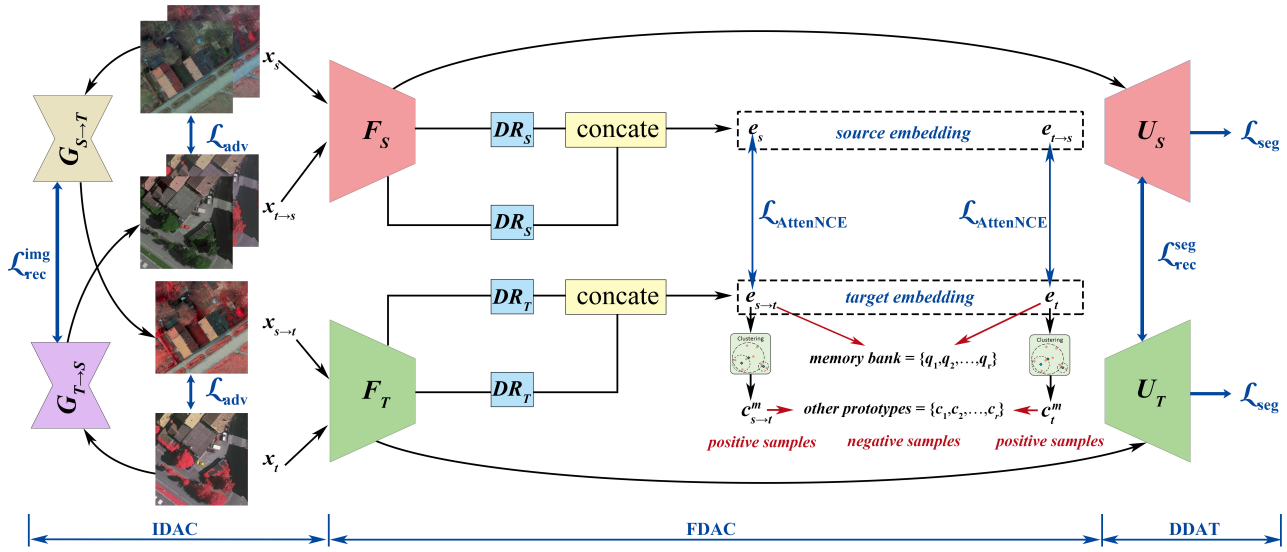


Fig. 2. MHDA framework. Our method is composed of three modules: IDAC, FDAC, and DDAT. IDAC is used for unsupervised unpaired heterogeneous data transformation. The input and translated images are then fed to their corresponding domain-specific encoders, and FDAC adopts the improved contrastive loss to extract the domain-invariant features for task predictions. Finally, DDAT uses the task consistent loss $\mathcal{L}_{\text{rec}}^{\text{seg}}$ to further regularize the DA training.

negative samples are not screened, it will inevitably cause new robustness problems. Therefore, we make the improvement based on the selective attention mechanism, as follows:

$$\mathcal{L}_{\text{Att-Info}} = -\log\left(\frac{\exp(e_s \cdot q_0/\tau)}{\sum_{i=0}^{r}\exp(e_s \cdot q_i \cdot \beta_i/\tau)}\right) \tag{9}$$

where β_i is the weight for screening the negative samples, as (10). Given that $e_{s\to t}$ denoted by q_0 is the positive sample of e_s, it does not need to be screened, thus $\beta_0 = 1$. But for the negative samples $\{q_i\}_{i=1}^{r}$, we calculate their *attention score* to the positive sample $e_{s\to t}$ with cosine similarity and reversed it to reduce the adverse effects brought by the similarity between the negative and positive samples

$$\beta_i = \begin{cases} 1, & i = 0 \\ 1 - \text{softmax}\left(\dfrac{e_{s\to t} \cdot q_i}{\|e_{s\to t}\| \cdot \|q_i\|}\right), & i \neq 0. \end{cases} \tag{10}$$

After that, with reference to prototypical contrastive learning (PCL) [45], we further avoid the above drawbacks using k-mean. Cluster the positive sample $e_{s\to t}$ M times with different numbers of clusters $L = \{l_m\}_{m=1}^{M}$, $l_m \gg r$. For the mth clustering operation, the cluster center c_0^m to which $e_{s\to t}$ belongs can be regarded as the positive prototype of e_s, and other randomly selected r cluster centers can be considered as the negative prototypes. With similarity measured by the dot product and NCE losses

$$\mathcal{L}_{\text{ProtoNCE}} = -\frac{1}{M}\sum_{m=1}^{M}\log\frac{\exp\left(e_s \cdot c_0^m/\phi_0^m\right)}{\sum_{i=0}^{r}\exp\left(e_s \cdot c_i^m/\phi_i^m\right)} \tag{11}$$

where ϕ_i^m is the level of concentration of embeddings around c_i^m, as follows:

$$\phi_i^m = \frac{\sum_{z=1}^{Z}\|e_z - c_i^m\|_2}{Z\log(Z+\alpha)} \tag{12}$$

where there are Z embeddings $\{e_z\}_{z=1}^{Z}$ belonging to the prototype c_i^m, and α is a smooth parameter.

Then, our contrastive loss $\mathcal{L}_{\text{AttenNCE}}$ for the feature-level constraint in the source domain can be calculated as follows:

$$\mathcal{L}_{\text{AttenNCE}}(X_S; F_S, F_T, G_{S\to T}) = \mathbb{E}_{x_s\sim X_S}(\mathcal{L}_{\text{Att-Info}} + \mathcal{L}_{\text{ProtoNCE}}). \tag{13}$$

Similarly, there is also the contrastive loss in the target domain $\mathcal{L}_{\text{AttenNCE}}(X_T; F_T, F_S, G_{T\to S})$.

H. Decision-Level Cross Domain Constraint $\mathcal{L}_{rec}^{seg}$

Given $\mathcal{T}_S = \mathcal{T}_T$ in the context of DA, there should be task consistency. Specifically, the prediction results of different domain patterns of the same instance should be consistent: $M_S(x_s) \approx M_T(x_{s\to t})$ and $M_S(x_{t\to s}) \approx M_T(x_t)$. The task consistent loss $\mathcal{L}_{\text{rec}}^{\text{seg}}$ can be used for further fine-grained domain alignment at the decision level, as follows:

$$\begin{aligned}\mathcal{L}_{\text{rec}}^{\text{seg}}(X_S, X_T; M_S, M_T, G_{S\to T}, G_{T\to S}) \\ = \mathbb{E}_{x_s\sim X_S}[\|M_S(x_s) - M_T(x_{s\to t})\|_1] \\ + \mathbb{E}_{x_t\sim X_T}[\|M_T(x_t) - M_S(x_{t\to s})\|_1].\end{aligned} \tag{14}$$

IV. Experimental Study

A. Domain Shift Criteria

For varied data sources of remote sensing images, whether there are domain shift problems that need to be addressed can be determined by the following four criteria.

Spatial resolution is defined by the pixel size of the imagery. A pixel or picture element is the smallest 2-D area that can be sensed by the remote sensing sensor. The higher the spatial resolution, the richer the object detail. Therefore, if there is a large difference in the spatial resolutions of the source and target domains, there will unavoidably be the domain shift in detail information, such as shape, texture, and geometric features.

Spectral resolution is typically defined as the number of portions of the electromagnetic spectrum in the remote sensing

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TABLE I
DOMAIN SHIFT ANALYSIS OF THE ISPRS, BIGCITY, AND WHU DATASETS. "=" MEANS THE SAME AND "≠" MEANS DIFFERENT

Dataset	Sub-dataset		Spatial Resolution		Spectral Resolution		Geographical Location		Imaging Condition	
			=	≠	=	≠	=	≠	=	≠
ISPRS	Vaihingen	Postdam		√		√		√		√
BigCity	MSS	Fusion		√		√	√			√
WHU	Aerial	Satellite II		√	√			√		√

sensor, and these portions are referred to as "bands." The more the bands, the higher the spectral resolution, the more the information, and the greater the interclass separability of some categories. If the training and test sets are constructed by remote sensing images with different numbers of bands, the information gap will inevitably result in the heterogeneous domain shift problem.

Geographical location is another important factor, given the vast observation range of remote sensing imagery. Typically, the same category has different patterns in different geographical locations, such as urban and rural buildings, and certain patterns are exclusive to certain geographical locations. However, it is impossible to obtain well-annotated samples that can encompass all the geographical patterns. Therefore, if the source and target domains are collected from different geographical locations, there will inevitably be nonnegligible differences in texture information, spatial position relations, etc., thus inducing domain shift problems.

Imaging condition is usually strongly correlated with the imaging time and date. The radiation intensity and spectral distribution of acquired images fluctuate with the light intensity and angle at different imaging times. And for some objects with obvious phenological changes, due to seasonal influence, their texture, spectrum, and other information vary substantially with different imaging dates. Therefore, there will inevitably be the domain shift between the source and target domains collected under different imaging conditions.

B. Dataset Description

There are three datasets for DA experiments, namely, the ISPRS, BigCity, and WHU datasets. Each of these datasets includes two sub-datasets, which are alternately used as the source and target domains in DA experiments on each dataset. Table I briefly describes the domain shift problems on each dataset, where the ISPRS and BigCity datasets involve heterogeneous domain shift, while the two sub-datasets of the WHU dataset are homogeneous. The details of each dataset are stated as follows.

1) ISPRS Dataset: This dataset for urban objects classification [63] contains two pixel-level annotated sub-datasets collected from two different geographical locations in Germany and annotates five major categories: *Impervious surfaces*, *Building*, *Low vegetation*, *Tree*, and *Car*.

Vaihingen dataset is collected from a small village with many single and small buildings and contains 33 aerial images with a size of about 2494 × 2064 pixels, where ten images are randomly selected for testing and the rest for training. And each image has a spatial resolution of 9 cm and contains three spectral bands: infra-red (IR), red (R), and green (G).

Postdam dataset is acquired from a historical city with large and dense buildings and narrow roads and provides 38 aerial images with a size of 6000 × 6000 pixels, which have a spatial resolution of 5 cm and contain a blue (B) band, in addition to IRRG. In DA experiments, there are 28 training images and ten test images.

2) BigCity Dataset: This dataset is constructed with GaoFen-2 satellite images and contains two sub-datasets, where five major categories are annotated: *Built-up*, *Road*, *Bare land*, *Vegetation*, and *Water*.

The GaoFen-2 satellite is equipped with two sensors. The panchromatic sensor acquires **PMS imagery** with 0.8-m spatial resolution and a panchromatic band (0.45–0.90 μm). The multispectral sensor collects **MSS imagery** with 3.24-m spatial resolution and four bands: B (0.45–0.52 μm), G (0.52–0.59 μm), R (0.63–0.69 μm), and IR (0.77–0.89 μm). **Fusion imagery** with 0.8-m spatial resolution and three bands of RGB is generated through sharpening and fusing the RGB bands of MSS imagery with the panchromatic band of PMS imagery.

MSS dataset collects 30 pixel-level annotated MSS images (IRRGB) with a size of 3000 × 3000 from five cities (Beijing, Tianjing, Shandong, Fujian, and Hainan) in China, including 20 training images and ten test images.

Fusion dataset contains 30 pixel-level annotated Fusion images (RGB) with a size of 3000 × 3000 from the same geographical location as MSS dataset, but their imaging time is different from each other. There are 20 images for training and ten images for testing.

3) WHU Dataset: This dataset [64] provides a large amount of pixel-level annotations for building extraction, including an aerial image sub-dataset and two satellite image sub-datasets. This article chooses two sub-datasets for DA experiments, namely, the Aerial and Satellite II datasets, respectively.

Aerial dataset contains 8189 aerial images collected from New Zealand with a size of 512 × 512 pixels, three bands of RGB, and a spatial resolution of 0.3 m, which contain various and versatile architecture types of buildings with different colors, sizes, and usages. In all, 5772 images are used for training, and the rest for testing.

Satellite II dataset is constructed by six neighboring satellite images covering 550 m^2 on East Asia with RGB bands and 2.7-m spatial resolution. And there are 17 388 pixel-level annotated patches with a size of 512 × 512 pixels, including 13 662 training images and 3726 test images.

C. Implementation

1) Network Architecture: DeepLabv3+ [65] is used as the baseline semantic segmentation network. $G_{S\to T}$ and $G_{T\to S}$

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are implemented as [22], where the input and output layers are adjusted according to the corresponding number of bands. D_S and D_T adopt four residual blocks [66] of which each convolution layer is followed by an InstanceNormalization layer and a ReLU activation layer. DR_S and DR_T consist of a GlobalAveragePooling layer, a fixed-dimensional (128-D) fully connected layer, and an $L2$-norm layer.

2) Training Detail: For each sub-dataset of the WHU dataset, the training samples are randomly flipped and rotated for data augmentation. For each sub-dataset of the ISPRS and BigCity datasets, the training set is augmented with one or more traditional data augmentation functions, as detailed in the supplementary material, to generate 10 000 patches with a size of 512 × 512 pixels randomly cropped from the augmented samples for model training.

After that, we implement our method with the TensorFlow framework and four 12 GB GPUs and use the stochastic gradient descent (SGD) as the optimizer, where the initial learning rate is 10^{-3}, the momentum is 0.9, and the weight decay is 10^{-6}, to train the model with a batch size of 2. Besides, some hyperparameters are set as follows: $\lambda_\theta = 0.999$ (2), $\lambda_{\mathrm{rec}}^{\mathrm{img}} = 10$ (5), $r = 2048$ (9) and (11), $\tau = 0.2$ (9), $m = 3$ (11), $L = \{3000, 6000, 9000\}$ (11), and $\alpha = 10$ (12).

3) Accuracy Evaluation Criteria: IoU (18) of each category and mIoU (19) are used as accuracy evaluation criteria on the ISPRS and BigCity datasets. The WHU dataset adopts IoU, precision (15), recall (16), and F_1 (17) to evaluate the effect of building extraction

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{15}$$

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{16}$$

$$F_1 = 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \tag{17}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FN}} \tag{18}$$

$$\mathrm{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{IoU}^i \tag{19}$$

where TP and TN are the number of positive and negative samples correctly predicted, respectively; FP and FN are the number of negative and positive samples incorrectly predicted, respectively. IoU^i is the IoU of the ith category; N is the number of categories.

D. Ablation Study

As aforementioned, MHDA implements heterogeneous unsupervised DA with three-level domain alignment modules. This section investigates the specific effect of each module in MHDA using the ISPRS and BigCity datasets, which involve heterogeneous domain shift problems. And the following four methods are used for ablation analysis: **1) baseline** that is trained only by the labeled data in the source domain, without any DA; **2) baseline + IDAC** that only implements CycleGAN at the instance level; **3) baseline + IDAC + FDAC** that used the contrastive loss for feature-level DA, in addition to the instance-level domain alignment with cycle consistency; and **4) baseline + MHDA** that further imposes the decision-level constraint based on task consistency for three-level domain alignment. Figs. 3–6 sketch their segmentation results under the optimal hyperparameters, and Tables V and VI show their recognition accuracy results.

TABLE II
INFLUENCE OF λ_{cyc} ON THE MIOU IN EACH DA EXPERIMENT

	0.1	0.2	0.5	1.0	2.0	5.0
Vaihingen→Postdam	0.3198*	0.3362	**0.3404**	0.3389	0.3101*	0.3056*
Postdam→Vaihingen	0.3532*	0.3608*	0.3695	**0.3766**	0.3656	0.3289*
MSS→Fusion	0.4518*	0.4572*	0.4677	**0.4685**	0.4656	0.4520*
Fusion→MSS	0.4795	0.4976	**0.5029**	0.4986	0.4913	0.4803

1) Baseline + IDAC: M_S, M_T, $G_{S\to T}$, $G_{T\to S}$, D_S, and D_T are jointly trained with $\mathcal{L}_{\mathrm{seg}} + \lambda_{\mathrm{cyc}}\mathcal{L}_{\mathrm{cyc}}$. Then, λ_{cyc} for each DA experiment can be determined by the best mIoU, which has been highlighted in bold in Table II. Whereas it is worth noting that improper λ_{cyc} even can make the baseline + IDAC worse than the baseline, namely, the values with $*$ in Table II.

After that, we analyze the generated images on each dataset and display the IRRG and RGB images of the real and synthetic data with four bands for better observation. In the Postdam → Vaihingen and MSS → Fusion image translations, comparing the synthetic images [see Figs. 4(c) and 5(c)] with the real images [see Figs. 3(b) and 6(a)] in the target domain, it can be found that IDAC can realize the style transfer that is the most obvious difference between the source and target domains. And in the Vaihingen → Postdam and Fusion → MSS image translations, aside from the style transfer on the shared bands of the two domains, there is an additional band in the source domain, as shown in Figs. 4(a) and 5(a), and IDAC also can effectively synthesize this additional band in the target domain, as shown in Figs. 3(c) and 6(c).

To investigate the effect of IDAC, the instance-level domain alignment module in MHDA, we compare the baseline + IDAC under the best λ_{cyc} with the baseline as follows. As shown in Tables V and VI, benefiting from the simulation ability of the synthetic images for image style differences, IDAC can improve the mIoU and the IoU of some categories to a certain extent, especially for *Low Vegetation* and *Tree* in the Postdam ⇄ Vaihingen experiments, as well as *Water* in the MSS ⇄ Fusion experiments. Nonetheless, it still has a negative impact on some category extraction at the same time (the values with $*$ in Tables V and VI). We argue that there are three causes of the negative domain alignment, i.e., the phenomenon in which the recognition accuracy after DA declines instead of increasing:

a) The limited simulation ability to the complex domain shift: The domain shift problems in remote sensing imagery exist not only in image style but also in other more complex factors, such as texture, shape, and spatial position relation. The cycle consistency loss $\mathcal{L}_{\mathrm{rec}}^{\mathrm{img}}$ used to maintain the image structure invariance makes the unsupervised unpaired image-to-image translation work out, but also limits the ability of the generated data to simulate the more complex domain shift.

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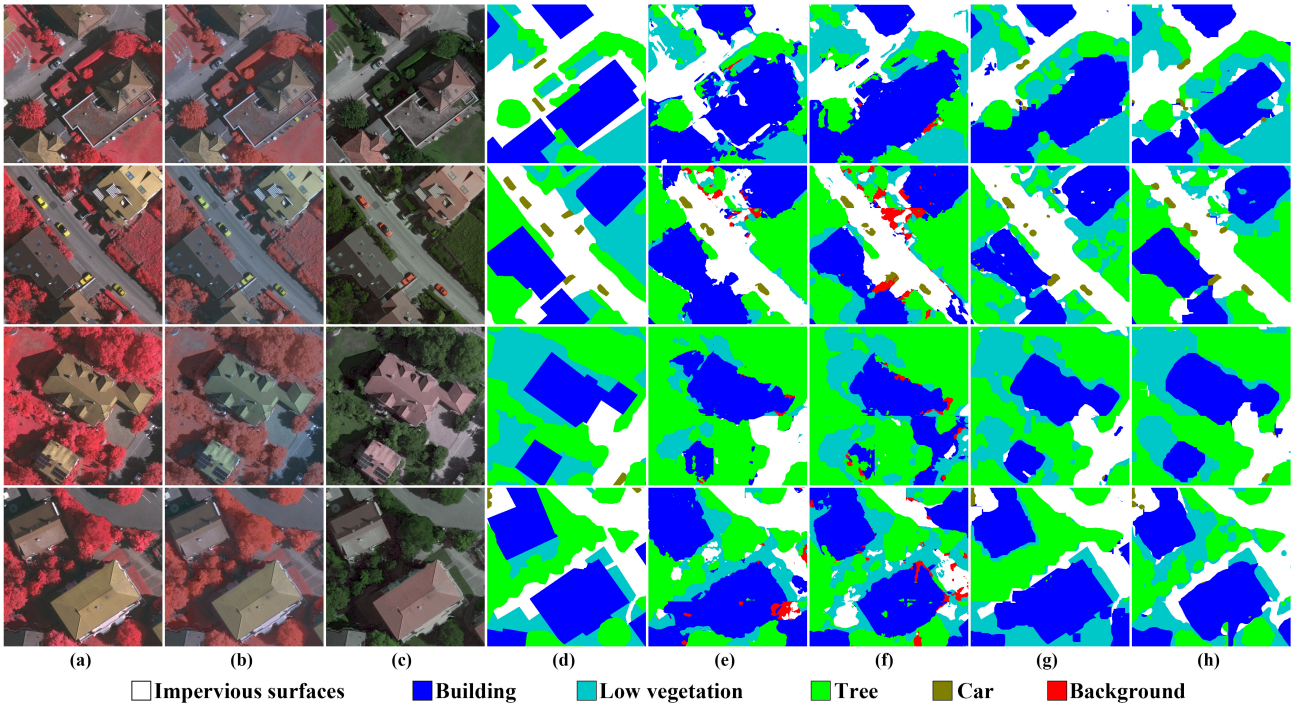


Fig. 3. Experimental results on the Vaihingen dataset. (a) IRRG original imagery. (b) IRRG generated imagery. (c) RGB generated imagery. (d) Ground truth. (e) Baseline. (f) Baseline + IDAC. (g) Baseline + IDAC + FDAC. (h) Baseline + MHDA.

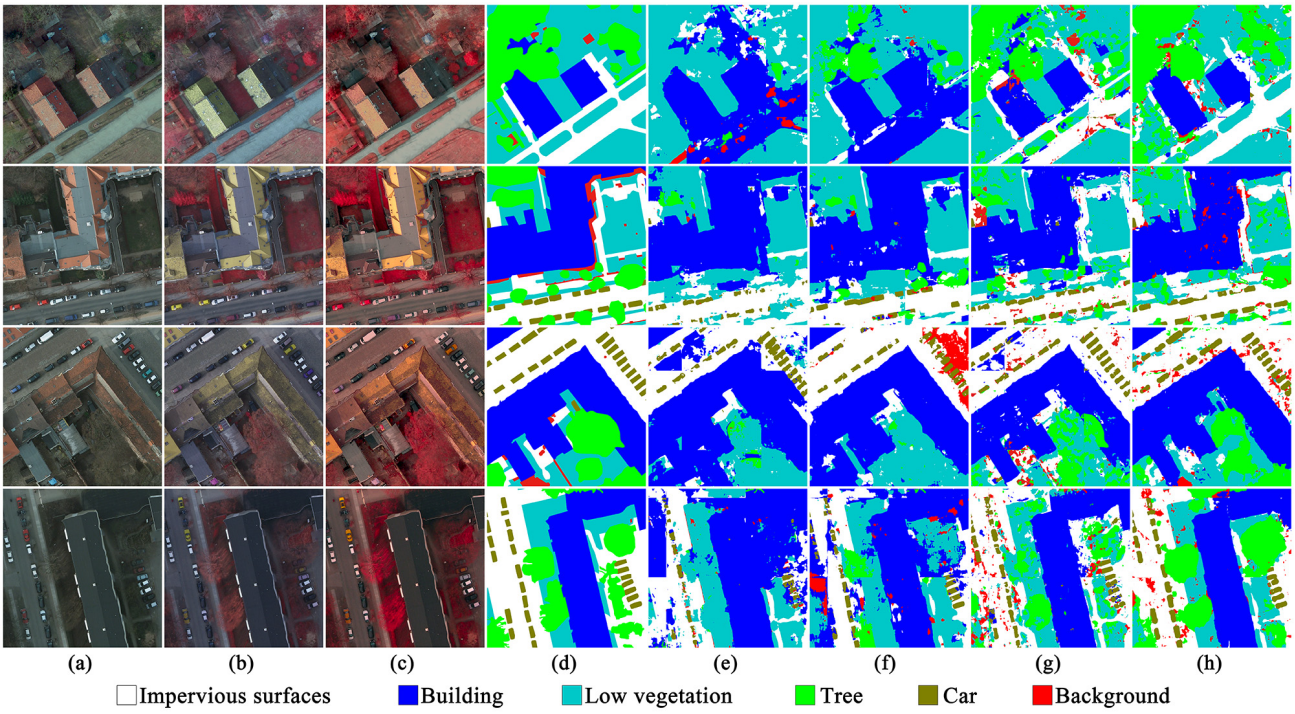


Fig. 4. Experimental results on the Postdam dataset. (a) RGB original imagery. (b) IRRG original imagery. (c) IRRG generated imagery. (d) Ground truth. (e) Baseline. (f) Baseline + IDAC. (g) Baseline + IDAC + FDAC. (h) Baseline + MHDA.

b) Subtle disturbances in synthetic images: Even if the image-to-image translation model has an excellent simulation ability in image style, there will inevitably be some subtle disturbances between the generated and real images. Especially after multiple high-dimensional linear functions in the CNN-based model, these disturbances are easy to cause completely contrary prediction results, as follows:

$$\begin{cases} x_t & = x_{s\to t} + \delta, \|\delta\|_\infty \leqslant \varepsilon \\ \omega^{\mathrm{T}} x_t & = \omega^{\mathrm{T}} x_{s\to t} + \omega^{\mathrm{T}} \delta \end{cases} \tag{20}$$

where δ is the subtle disturbance; ω is the parameter of the CNN-based model.

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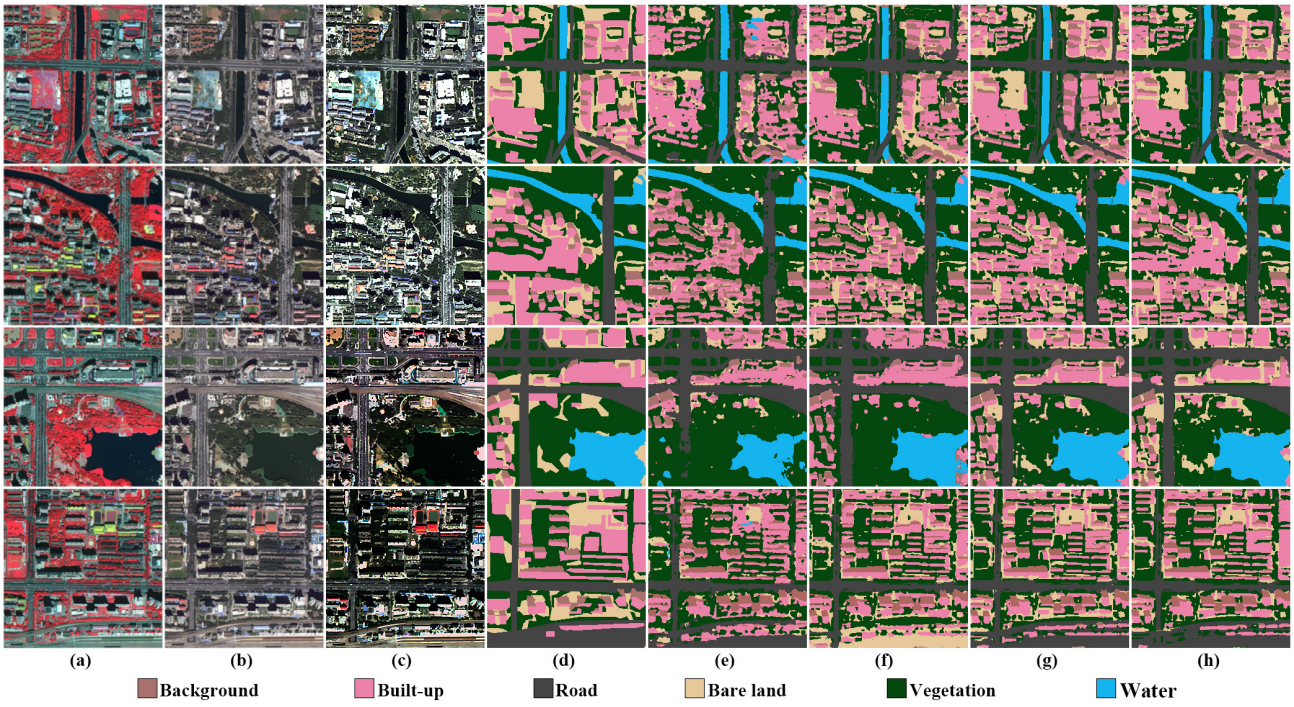


Fig. 5. Experimental results on the MSS dataset. (a) IRRG original imagery. (b) RGB original imagery. (c) IRRG generated imagery. (d) Ground truth. (e) Baseline. (f) Baseline + IDAC. (g) Baseline + IDAC + FDAC. (h) Baseline + MHDA.

TABLE III
INFLUENCE OF λ_{ATT} ON THE MIOU IN EACH DA EXPERIMENT

	0.1	0.5	1	2	5	10
Vaihingen→Postdam	0.4310	0.4380	0.4413	0.4610	**0.4699**	0.4363
Postdam→Vaihingen	0.4483	0.4574	0.4667	**0.4733**	0.4577	0.4229
MSS→Fusion	0.4776	0.5319	**0.5521**	0.5426	0.5237	0.4994
Fusion→MSS	0.5213	0.5581	0.5609	**0.5726**	0.5595	0.5292

TABLE IV
INFLUENCE OF λ_{rec}^{seg} ON THE MIOU IN EACH DA EXPERIMENT

	0.1	0.5	1	2	5	10
Vaihingen→Postdam	0.4746	0.4799	**0.4963**	0.4927	0.4831	0.4725
Postdam→Vaihingen	0.4873	0.4892	0.4995	**0.5051**	0.4826	0.4795
MSS→Fusion	0.5588	0.5702	0.5821	**0.5915**	0.5772	0.5694
Fusion→MSS	0.5782	0.5831	0.5909	**0.6004**	0.5894	0.5799

c) The insufficient attention to the real data in the target domain: The real unlabeled target domain data only participate in the training of CycleGAN, which makes the cognition of M_T to the target domain all come from the pseudosamples synthesized from the source domain. Ignoring the real data distribution in the target domain is also an important reason for the negative domain alignment.

2) Baseline + IDAC + FDAC: F_T (except for the first convolution layer) and DR_T are momentum updated, and other modules are updated with $\mathcal{L}_{seg}+\lambda_{cyc}\mathcal{L}_{cyc}+\lambda_{att}\mathcal{L}_{AttenNCE}$. We determine λ_{att} for each DA experiment based on the best mIoU, which has been emphasized in bold in Table III.

On the basis of the best λ_{att} and λ_{cyc}, the DA effect of FDAC, the feature-level domain alignment module in MHDA, is further explored by comparing the baseline + IDAC + FDAC with the baseline + IDAC. In the Vaihingen $\rightleftarrows$ Postdam experiments, as shown in Table V, mIoU is improved by about 10.0%. *Impervious surfaces*, *Building*, and *Car* are all greatly improved, which means that FDAC can effectively deal with more complex differences in shape and geometric information. Besides, for *Low vegetation* and *Tree*, of which the domain shift problems exist not only in image style but also in texture features due to different spatial resolutions, their IoU can still be significantly ameliorated by FDAC, which also illustrates that FDAC can further tackle the domain shift in texture information.

In the MSS $\rightleftarrows$ Fusion experiments, the huge difference in the spatial resolution also brings about the great domain shift in shape and texture information for each category. After FDAC, as shown in Table VI, the IoU of each category has been greatly improved, and mIoU has grown by about 8.5% and 7.0%, respectively, which further highlights the ability of FDAC to handle the more complex domain shift. Furthermore, it should be noted that there is no negative domain alignment phenomenon in any category after using FDAC, which also demonstrates that it can effectively eliminate the noise interference in synthetic images generated in IDAC.

3) Baseline + MHDA: The modules in MHDA are jointly trained according to the parameter update method described in Section III-D. λ_{rec}^{seg} for each DA experiment is determined by the best mIoU, namely, the bold values in Table IV.

Based on the best λ_{att}, λ_{cyc}, and λ_{rec}^{seg}, the following compares the baseline + MHDA with the baseline + IDAC + FDAC to further investigate the DA effect of DDAT, namely, the decision-level domain alignment module in MHDA. As shown in Table V, in the Vaihingen → Postdam experiment, DDAT

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TABLE V
CLASSWISE IOU OF DIFFERENT METHODS IN TWO DA EXPERIMENTS ON THE ISPRS DATASET

	Vaihingen→Postdam						Postdam→Vaihingen					
	IS	Building	LV	Tree	Car	$mIoU$	IS	Building	LV	Tree	Car	$mIoU$
baseline	0.4768	0.3823	0.2900	0.0972	0.3656	0.3224	0.3546	0.5676	0.2331	0.4586	0.1941	0.3616
baseline+UDA	0.3564*	0.2551*	0.3277	0.1093	0.3688	0.2834*	0.3872	0.5097*	0.2411	0.4731	0.1068*	0.3436*
baseline+AdaptSegNet	0.4826	0.4269	0.2609*	**0.2254**	0.3995	0.3591	0.5034	**0.6618**	**0.2839**	0.5594	0.0984*	0.4214
baseline+CyCADA	0.4963	0.5268	**0.3461**	0.1151	0.4901	0.3949	0.5207	0.6286	0.2653	0.5488	0.2404	0.4408
baseline+MDA	**0.6299**	**0.5769**	0.3331	0.1630	**0.5106**	**0.4427**	**0.5549**	0.6560	0.2766	**0.5619**	**0.3298**	**0.4759**
baseline+IDAC	0.4914	0.3852	0.3191	0.1267	0.3795	0.3404	0.3564	0.5744	0.2584	0.4869	0.2068	0.3766
baseline+IDAC+FDAC	0.6327	0.5845	0.3779	0.2129	0.5299	0.4699	0.5387	0.6410	0.3157	0.5767	0.2942	0.4733
baseline+MHDA	0.6599	0.6096	0.4040	0.2525	0.5555	0.4963	0.5763	0.6744	0.3226	0.6047	0.3476	0.5051

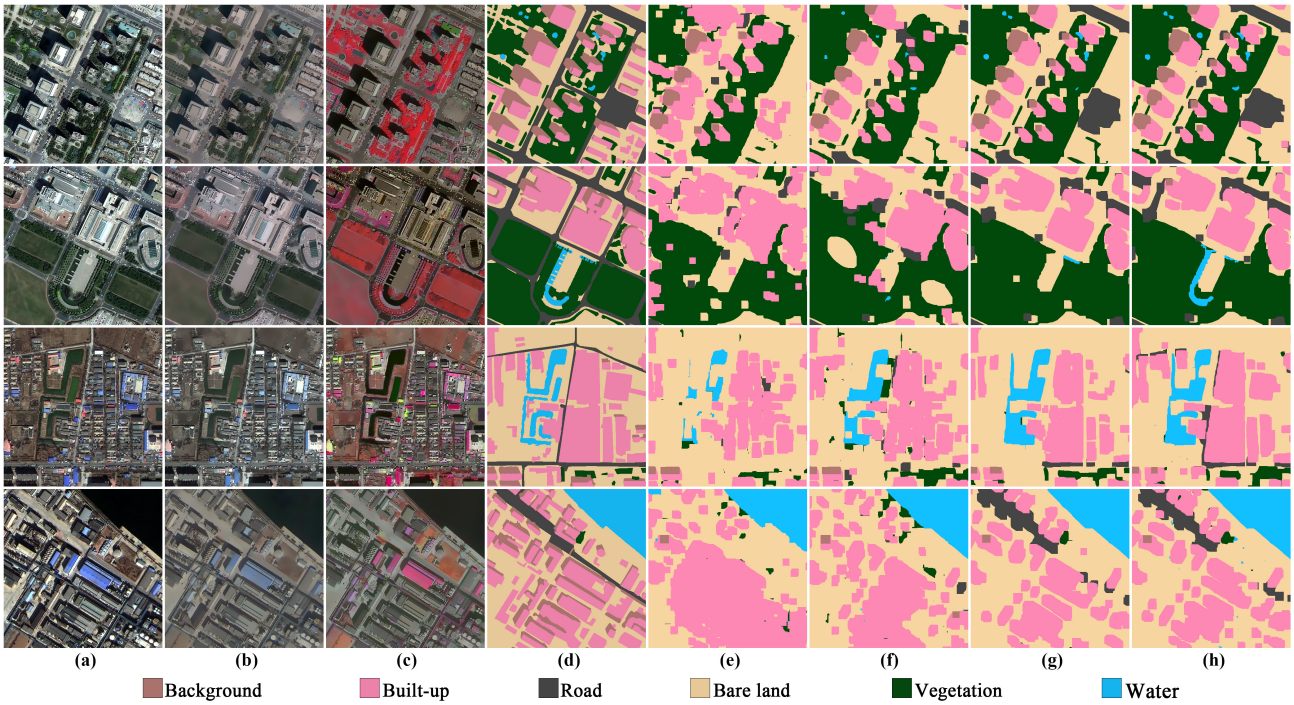


Fig. 6. Experimental results on the Fusion dataset. (a) RGB original imagery. (b) RGB generated imagery. (c) IRRG generated imagery. (d) Ground truth. (e) Baseline. (f) Baseline + IDAC. (g) Baseline + IDAC + FDAC. (h) Baseline + MHDA.

improves the mIoU and the IoU of each category by above 2.5%, which indicates that the task consistent loss facilitates further improving the DA effect. In the Postdam → Vaihingen experiment, DDAT has greatly improved the extraction of some categories with distinct boundaries, such as *Impervious surfaces*, *Building*, and *Car*. Especially for *Car*, the small-scale category, its IoU grows by nearly 5.5% after using DDAT.

As shown in Table VI, in the MSS → Fusion experiment, its mIoU can be improved by about 4.0% after DDAT. The improvement of *Road* is the most remarkable (>7.5%), and as shown in Fig. 6(h), DDAT can significantly promote the extraction of some narrow roads. In the Fusion → MSS experiment, DDAT improves the mIoU by about 3.0%. The IoU of *Built-up* and *Bare land* increases by about 3.5%, and the improvement of other categories also reaches to about 2.5%. Furthermore, intuitively, as shown in Fig. 5(h), the structural integrity of each category has also been ameliorated to varied degrees after DDAT.

E. Comparison Study

For fairer comparisons, our homogeneous DA method, termed MDA, is implemented, which differs only from MHDA in that its IDAC only achieves the homogeneous data transformation. This section investigates the necessity of heterogeneous DA by comparing MHDA with MDA and explores the effectiveness of multilevel DA through the advantages of MDA over other state-of-the-art homogeneous DA methods. It is worth noting that for the ISPRS and BigCity datasets with heterogeneous domain shift problems, all the homogeneous DA methods are performed on the shared bands, i.e., Vaihingen IRRG ⇄ Postdam IRRG and MSS RGB ⇄ Fusion RGB.

1) MHDA Versus MDA: To explore the necessity of heterogeneous DA and assess the role of the additional band information, we compare MHDA with MDA through DA experiments on the ISPRS and BigCity datasets, as shown in Tables V and VI, and the specific analysis is as follows.

a) Vaihingen → Postdam: MHDA improves the mIoU by about 5.5%. *Low vegetation* and *Tree* are significantly

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TABLE VI
CLASSWISE IOU OF DIFFERENT METHODS IN TWO DA EXPERIMENTS ON THE BIGCITY DATASET

	MSS→Fusion						Fusion→MSS					
	Built-up	Road	Bare land	Vegetation	Water	$mIoU$	Built-up	Road	Bare land	Vegetation	Water	$mIoU$
baseline	0.4245	0.3150	0.5593	0.4097	0.6007	0.4618	0.4082	0.5056	0.3208	0.4829	0.6402	0.4715
baseline+UDA	0.4374	0.1813*	0.4212*	0.4340	0.6451	0.4238*	0.3979*	0.4917*	0.2048*	0.4516*	0.7019	0.4496*
baseline+AdaptSegNet	0.5489	0.4365	0.5322*	0.4089	0.6813	0.5215	0.5009	0.5304	0.3841	0.4875	0.6704	0.5147
baseline+CyCADA	0.5299	0.3622	0.4985*	0.4704	0.7043	0.5131	0.4665	0.5833	0.3329	0.5075	0.7053	0.5191
baseline+MDA	**0.5668**	**0.4672**	**0.5961**	**0.4756**	**0.7190**	**0.5649**	**0.5129**	**0.6089**	**0.4085**	**0.5087**	**0.7206**	**0.5519**
baseline+IDAC	0.4465	0.3346	0.5292*	0.3735*	0.6588	0.4685	0.4589	0.5205	0.3005*	0.4826*	0.7018	0.4929
baseline+IDAC+FDAC	0.5542	0.4028	0.5834	0.5046	0.7154	0.5521	0.5002	0.5976	0.4148	0.5809	0.7193	0.5626
baseline+MHDA	0.5831	0.4788	0.6112	0.5431	0.7414	0.5915	0.5336	0.6206	0.4488	0.6049	0.7443	0.5904

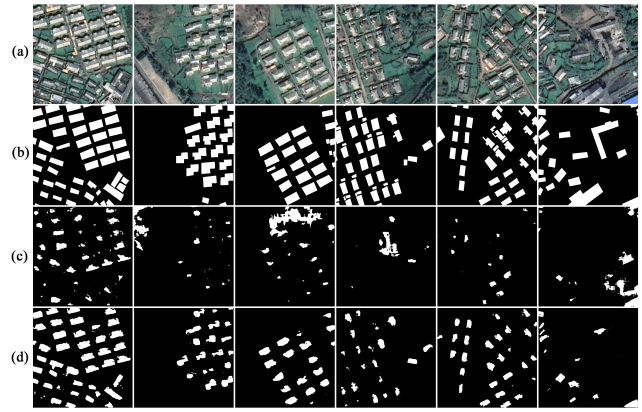


Fig. 7. Segmentation results in the Aerial → Satellite II experiment. (a) Imagery. (b) Ground truth. (c) Baseline. (d) Baseline + MDA.

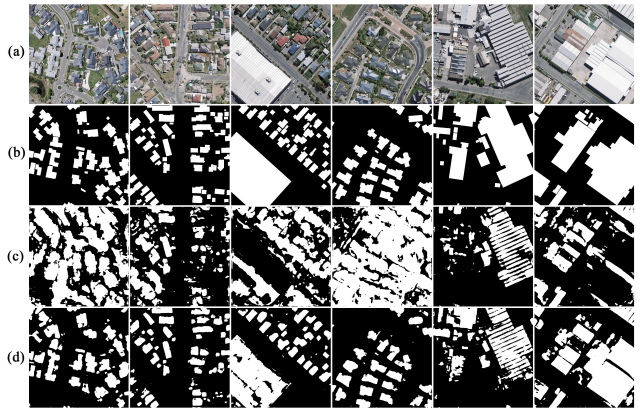


Fig. 8. Segmentation results in the Satellite II → Aerial experiment. (a) Imagery. (b) Ground truth. (c) Baseline. (d) Baseline + MDA.

improved by greater than 7.0%, and the improvement of other categories can also reach to 3.0%–4.5%.

b) Postdam → Vaihingen: The improvement of mIoU brought by MHDA is about 3.0%. *Low vegetation* and *Tree* still have the greater increase (>4.0%) than other categories.

c) MSS → Fusion: Its mIoU increases by above 2.5%. Besides, *Vegetation* is improved by above 6.5%, and the improvement of other categories can also reach to 1.0%–2.5%.

d) Fusion → MSS: MHDA still brings the most significant improvement (>9.5%) to *Vegetation*, and *Bare land* is also greatly improved (>4.0%). Furthermore, the overall accuracy mIoU grows nearly 4.0%.

To sum up, in each DA experiment, the additional band information has a great influence on some categories with the domain shift problems owing more to the differences in spectral information, such as *Low vegetation* and *Tree* in the Postdam ⇄ Vaihingen experiments, as well as *Vegetation* in the MSS ⇄ Fusion experiments. Besides, MHDA still outperforms MDA on other categories, although the improvement is not as significant as that of the aforementioned categories. Therefore, we conclude that benefiting from the additional band information, MHDA has a better performance on the overall accuracy and recognition accuracy of each category than MDA, which demonstrates the necessity of heterogeneous DA in the remote sensing image segmentation task.

2) MDA Versus Other Methods: We conduct comparative experiments between MDA and other state-of-the-art homogeneous DA methods mentioned in Section II-G on the ISPRS, BigCity, and WHU datasets, and the segmentation results of MDA on the WHU dataset are exhibited in Figs. 7 and 8. Tables V–VII show their recognition accuracy results, where the optimal values are highlighted in bold, the suboptimal values are underlined, and the values worse than the baseline are marked with ∗. The following will specifically analyze the DA effect of each homogeneous DA method through its improvement over the baseline to investigate the effectiveness of multilevel DA.

a) UDA: In the Aerial ⇄ Satellite II experiments, many accuracy criteria decline after UDA, and the growth of other accuracy criteria is also limited. In DA experiments on the ISPRS and BigCity datasets, mIoU all declines rather than increases after UDA, and the decreases are even close to 4.0% in the Postdam → Vaihingen and MSS → Fusion experiments. However, it still has certain potential in the extraction of several categories, such as *Low vegetation* in the Vaihingen → Postdam experiment, *Impervious surfaces* in the Postdam → Vaihingen experiment, and *Water* in the MSS ⇄ Fusion experiments. Overall, the DA effect of UDA is limited, and it only implements instance-level DA with CycleGAN, which results in many causes of the negative domain alignment phenomenon, such as the inability to simulate the more complex domain shift, and some subtle disturbances that can be infinitely amplified by high-dimensional neural networks.

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TABLE VII
RECOGNITION ACCURACY OF DIFFERENT METHODS IN TWO DA EXPERIMENTS ON THE WHU DATASET

	Aerial → Satellite II				Satellite II → Aerial			
	$precision$	$recall$	F_1	IoU	$precision$	$recall$	F_1	IoU
baseline	0.8665	0.0714	0.1319	0.0706	0.6393	0.3397	0.4437	0.2851
baseline + UDA	0.5624*	0.0942	0.1613	0.0877	0.6635	0.2987*	0.4120*	0.2594*
baseline + AdaptSegNet	**0.9220**	0.3473	0.5045	0.3373	0.8280	0.4089	0.5474	0.3768
baseline + CyCADA	0.7882*	0.3518	0.4865	0.3214	0.7810	0.5116	0.6182	0.4474
baseline + MDA	0.8734	**0.4735**	**0.6141**	**0.4431**	**0.8477**	**0.6591**	**0.7416**	**0.5893**

b) AdaptSegNet: As a DA method that only adopts decision-level DA, its performance is more stable and excellent than UDA. In DA experiments on the WHU dataset, AdaptSegNet provides significant improvement for each accuracy criterion, a handful of accuracy criteria reach to the optimal values, and $precision$ even exceeds to that of MDA in the Aerial → Satellite II experiment. In DA experiments on the ISPRS and BigCity datasets, although AdaptSegNet still has a negative impact on some categories, its mIoU is second only to MDA in the MSS → Fusion experiment, and the improvement of some categories even exceeds MDA in the Vaihingen ⇄ Postdam experiments. Therefore, the following inferences can be drawn: 1) decision-level DA has noticeable advantages over instance-level DA in remote sensing image segmentation and 2) AdaptSegNet can provide competitive improvement for some categories compared with other DA methods, but it still suffers from negative domain alignment due to the absence of the real label space in the target domain.

c) CyCADA: The combination of instance-level DA and feature-level DA makes CyCADA second only to MDA in most DA experiments. While AdaptSegNet outperforms CyCADA on the mIoU of the MSS → Fusion experiment and the precision of the Aerial ⇄ Satellite II experiments, which further indicates the effectiveness of decision-level DA. And with the aid of further feature-level domain alignment, CyCADA has noticeable advantages over UDA, but the IoU of *Bare land* in the MSS → Fusion experiment and the precision of the Aerial → Satellite II experiment are still inferior to those of the baseline, which demonstrates that the feature-level DA adopted by CyCADA contributes to avoiding some noise interference brought by its instance-level DA, but the improvement ability is not enough to completely get rid of the negative domain alignment phenomenon.

d) MDA: Even though MDA performs worse on several categories than AdaptSegNet or CyCADA in the Vaihingen ⇄ Postdam experiments and is inferior to AdaptSegNet on the precision of the Aerial → Satellite II experiment, the strategy of unifying the three-level DA into a framework can effectively eliminate the negative domain alignment phenomenon and yield superior accuracy results in each DA experiment. It suggests that MDA is a promising solution for the more effective fine-grained domain alignment in the remote sensing image segmentation task, attributed to the complementary advantages of multilevel DA.

In conclusion, UDA that only adopts instance-level DA is prone to the negative domain alignment phenomenon. AdaptSegNet only uses decision-level DA, and CyCADA combines instance-level and feature-level DA, which can achieve better DA effects than UDA, but the performance on some categories is still unsatisfactory. MDA can make full use of the advantages of each-level DA and realize the best DA effect in each DA experiment. Therefore, comparing MDA with other homogeneous methods, we conclude that the multilevel DA proposed by us is more suitable for the fine-grained remote sensing image segmentation task.

V. CONCLUSION

To tackle the more complex domain shift problems and fully use heterogeneous data information without the label information from the target domain, we propose a novel multilevel heterogeneous unsupervised DA framework for the remote sensing image segmentation task, termed MHDA, and conduct extensive DA experiments on the ISPRS, BigCity, and WHU datasets. Through ablation and comparison studies, we explore the DA effect of each module in MHDA, including IDAC, FDAC, and DDAT, and investigate the necessity of heterogeneous DA and the effectiveness of multilevel DA. The following conclusions can be drawn.

1) IDAC can simultaneously realize image style transfer and synthesize the additional band for effective unsupervised unpaired heterogeneous data transformation.
2) FDAC can extract the domain-invariant features with better generalization ability and effectively suppress the noise interference of synthetic data generated in IDAC.
3) DDAT can improve the DA effect with the further fine-grained constraint at the decision level, especially for some small-scale categories.
4) Heterogeneous DA can further mine the additional band information, which facilitates some category extraction, especially for those categories with the domain shift owing more to the differences in spectral information.
5) Multilevel DA can effectively exploit the complementary advantages of each-level DA and handle the more complex domain shift problems in remote sensing imagery.

Although MHDA realizes the prominent DA effect, the real data X_T in the target domain are generally not accessible as well in the real world. In response to this problem, in recent years, several research has already started to concentrate on domain generalization (DG), which attempts to make the model generalize well to the unseen target domain by the use of several different source domains. And in the future, we will

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be further devoted to the effective DG method for the remote sensing image segmentation task.

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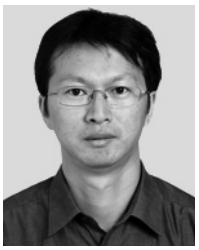
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